

PHYUGMCC1101/MIN1101: Mechanics::Sample Questions, Problems with Hints/Solution

1. State and prove Work-energy theorem. [**Statement:** The principle of work and kinetic energy (also known as the work–energy theorem) states that the work done by all forces acting on a particle (the work of the resultant force) equals the change in the kinetic energy of the particle. That is, the work W done by the resultant force on a particle equals the change in the particle's kinetic energy, $W = \int_i^f \vec{F} \cdot d\vec{r} = \frac{1}{2}mv_i^2 - \frac{1}{2}mv_f^2 = \Delta E_k$. **Proof:** $W = \int_i^f \vec{F} \cdot d\vec{r} = \int_i^f \vec{F} \cdot \vec{v} dt = \int_i^f F v dt = \int_i^f (ma) v dt = \int_i^f m \left(\frac{dv}{dt} \right) v dt = m \int_i^f v dv = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \Delta E_k = \text{Change in kinetic energy.}]$

2. What are conservative and non-conservative forces. Give examples. [**Conservative forces:** The forces, whose total work is path independent, are called conservative forces. Example: Gravitational force, electrostatic force, spring force, etc. Such forces are derivable from a scalar potential (say, U) ($F = -\nabla U$). Under conservative force work done in a round trip becomes zero.

Non-conservative forces: Non-conservative forces are dissipative forces such as friction or air resistance. These forces take energy away from the system as the system progresses, energy that you can't get back. These forces are path dependent; therefore it matters where the object starts and stops. Under non-conservative force work done in a round trip is non-zero. Additional Note: The work done by a conservative force is $W = -\Delta U$, where ΔU is the change in the potential energy associated with the force. The negative sign provides the convention that work done against a force field increases potential energy, while work done by the force field decreases potential energy.

If the force acting on a body varies over space, then one has a force field; such a field is described by vectors at every point in space, which is in-turn called a vector field. A conservative vector field can be simply expressed as the gradient of a certain scalar function, called a scalar potential.]

3. What do you understand by central force and non-central force? Give examples of central force and non-central force. [**Central force:** In classical mechanics¹, a central force on an object is a force that is directed towards or away from a point called center of force. $\vec{F} = \mathbf{F}(\mathbf{r}) = F(r)\hat{r}$, where $\vec{F}$ is the force, $\mathbf{F}$ is a vector valued force function, F is a scalar valued force function, $\mathbf{r}$ is the position vector, r is its length. Examples: Gravitational force, Coulomb force, Elastic force in spring, etc.]

Non-central force: Non-central force is a force between two particles that is not directed along the line connecting them. Example: Spin-dependent nuclear forces, magnetic force between two current carrying loops.

4. Obtain expressions for the radial and transverse (tangential) components of velocity and acceleration in plane polar coordinate system. [Start with:

¹Classical mechanics is a physical theory describing the motion of objects such as projectiles, parts of machinery, spacecraft, planets, stars, and galaxies.

$\vec{r} = r\hat{r}$, use: $\frac{d\hat{r}}{d\theta} = \hat{\theta}$, $\frac{d\hat{\theta}}{d\theta} = -\hat{r}$; $\frac{d\hat{r}}{dt} = \dot{\theta}\hat{\theta}$, $\frac{d\hat{\theta}}{dt} = -\dot{\theta}\hat{r}$.

Velocity, $\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{r}) = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt} = \dot{r}\hat{r} + r\frac{d\hat{r}}{d\theta}\frac{d\theta}{dt} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} = v_r\hat{r} + v_\theta\hat{\theta}$. Giving: Radial component of velocity, $v_r = \dot{r}$, and Tangential (or Transverse) component of velocity, $v_\theta = r\dot{\theta}$.

Acceleration: $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{r} + r\dot{\theta}\hat{\theta}) = \frac{d}{dt}(\dot{r}\hat{r}) + \frac{d}{dt}(r\dot{\theta}\hat{\theta}) = (\frac{d\dot{r}}{dt}\hat{r} + \dot{r}\frac{d\hat{r}}{dt}) + (\frac{dr}{dt}\dot{\theta}\hat{\theta} + r\frac{d\dot{\theta}}{dt}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt}) = (\ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta}) + (\dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}(-\dot{\theta}\hat{r})) = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} = a_r\hat{r} + a_\theta\hat{\theta}$. Given: Radial component of acceleration, $a_r = \ddot{r} - r\dot{\theta}^2$, and Tangential (or Transverse) component of acceleration, $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$

- Write down the equation of motions of a system of two-particles under the action of a central force. Now reduce the two-body central force problem to a one-body problem. [Let F_{12} is the force on m_1 due to m_2 , $F_{12} = F(r)\hat{r}$, and that on m_2 due to m_1 will be equal and opposite, $F_{21} = -F(r)\hat{r}$. $\therefore m_1 \frac{d^2 r_1}{dt^2} = F_{12} = F(r)\hat{r}$; $m_2 \frac{d^2 r_2}{dt^2} = F_{21} = -F(r)\hat{r}$; $\frac{d^2 r_1}{dt^2} - \frac{d^2 r_2}{dt^2} = (1/m_1 + 1/m_2)F(r)\hat{r} \Rightarrow \frac{d^2 r}{dt^2} = \mu F(r)\hat{r} \Rightarrow \mu \ddot{r} = F(r)\hat{r}$. Here $r = r_1 - r_2$ = relative coordinate. This is the equation of motion of CM of a two-body system. In this way two-body problem can be reduced to a one-body problem.]
- Give examples of inverse square law forces. Show that inverse square law forces are central forces. Check whether such forces are conservative or not.
- Prove the law of conservation of energy for an object moving under the action of a central force. Or, Prove that $\frac{1}{2}mv^2 + U + \frac{L^2}{2mr^2} = E = \text{Constant}$, where symbols have their usual meanings.
- Discuss the nature of path of particle moving under the action of central force. [Hints: We have the total energy, $E = \frac{1}{2}mv^2 + U + \frac{L^2}{2mr^2} = \frac{1}{2}mv^2 + \frac{L^2}{2mr^2} \pm \frac{c}{r} = \frac{1}{2}mv^2 + U'$; $U' = \frac{L^2}{2mr^2} \pm \frac{c}{r}$. For discussion on the nature of path let us rewrite the above relation as:

$$\frac{1}{2}mv^2 = E - U'$$

- 1) In the classically forbidden region, kinetic energy will be negative, since kinetic energy can never be negative in the classically allowed region. So, accepted value of total energy must satisfy the criteria: $E \geq U'_{min}$.
- (2) At the classical turning point the kinetic energy, $\frac{1}{2}mv^2 = E - U'(r_{tp}) = 0$. When, $E = U'_{min}$, there will be only one turning point and the path will be circular. So, particle under attractive inverse square law of force, the orbit will be circular.
- 3) If for some energy, $E = U'(r_1) = U'(r_2)$, then, there will be two turning points and the path will be an ellipse, with perihelion (r_1) and aphelion (r_2).
9. Show that the differential equation of motion of a particle of mass m under the control of a central isotropic force can be written as

$$\frac{d^2 u}{d\theta^2} + u = -\frac{m}{L^2 u^2} F\left(\frac{1}{u}\right)$$

[Solution: For motion under the action of a central force, we know that the tangential component of particle acceleration is zero i.e., $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0$

$2(\dot{r})\dot{\theta} = 0, \Rightarrow \frac{d}{dt}(mr^2\dot{\theta}) = 0, \Rightarrow mr^2\dot{\theta} = L = \text{constant}$. Hence, $\dot{\theta} = \frac{L}{mr^2}$. Now, the radial component of force, $F(r) = ma_r = m\ddot{r} - mr\dot{\theta}^2$. Let $r = \frac{1}{u}$, $\dot{r} = \frac{d}{du}(\frac{1}{u}) \frac{du}{d\theta} \frac{d\theta}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} (Lu^2/m) = -\frac{L}{m} \frac{du}{d\theta}$; $\ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{d}{dt}(-\frac{L}{m} \frac{du}{d\theta}) = \frac{d}{d\theta}(-\frac{L}{m} \frac{du}{d\theta}) \frac{d\theta}{dt} = (-\frac{L}{m} \frac{d^2u}{d\theta^2})(Lu^2/m) = -\frac{L^2u^2}{m^2} \frac{d^2u}{d\theta^2}$; $F(r) = m\ddot{r} - mr\dot{\theta}^2 = m(-\frac{L^2u^2}{m^2} \frac{d^2u}{d\theta^2}) - m(\frac{1}{u})(Lu^2/m)^2 = -\frac{L^2u^2}{m}(\frac{d^2u}{d\theta^2} + u) \Rightarrow$

$$F(1/u) = -\frac{L^2u^2}{m}(\frac{d^2u}{d\theta^2} + u)$$

Required law of force.]

10. The path of a particle moving under central force field is represented by $r = a \sin \theta$. Find the corresponding formula for the force law. [Solution: $r = a \sin \theta, \Rightarrow u = \frac{1}{r} = \frac{1}{a \sin \theta} = \frac{\text{cosec} \theta}{a}$, $\frac{du}{d\theta} = -\frac{\text{cosec} \theta \cot \theta}{a}$, $\frac{d^2u}{d\theta^2} = -\frac{d}{d\theta}(\frac{\text{cosec} \theta \cot \theta}{a}) = -\frac{(-\text{cosec} \theta \cot \theta) \cot \theta + \text{cosec} \theta (-\text{cosec}^2 \theta)}{a} = \frac{\text{cosec} \theta (\cot^2 \theta + \text{cosec}^2 \theta)}{a} = \frac{\text{cosec} \theta}{a} = \frac{1}{a \sin \theta} = u$; $\therefore F(r) = F(1/u) = -\frac{L^2u^2}{m}(\frac{d^2u}{d\theta^2} + u) = -\frac{L^2u^2}{m}(u + u) = -\frac{L^2u^2}{m}(2u) = -\frac{2L^2u^3}{m} = -\frac{2L^2}{mr^3}$; $F(r) \propto -\frac{1}{r^3}$]
11. Starting from the differential equation of motion of a particle moving under central force:

$$F(r) = F(1/u) = -\frac{L^2u^2}{m}(\frac{d^2u}{d\theta^2} + u)$$

obtain the polar (r, θ) equation of the orbit of a particle moving under an inverse square law of attractive force (eg., gravitational force). [Solution: For an inverse square law of attractive force, like the force of gravitation, we have $F(r) = -\frac{GMm}{r^2} = -\frac{c}{r^2} = -cu^2$. $\therefore \frac{d^2u}{d\theta^2} + u = -\frac{m}{L^2u^2}F(1/u) = -\frac{m}{L^2u^2}(-cu^2) = \frac{mc}{L^2}$; $\Rightarrow \frac{d^2u}{d\theta^2} + u - \frac{mc}{L^2} = 0 \Rightarrow \frac{d^2}{d\theta^2}(u - \frac{mc}{L^2}) + (u - \frac{mc}{L^2}) = 0 \Rightarrow (u - \frac{mc}{L^2}) = A \cos(\theta - \theta_0) \Rightarrow u = \frac{mc}{L^2} + A \cos(\theta - \theta_0) = \frac{mc}{L^2}(1 + \frac{AL^2}{mc}) \cos(\theta - \theta_0) \Rightarrow \frac{1}{r} = \frac{mc}{L^2}(1 + \frac{AL^2}{mc}) \cos(\theta - \theta_0)$. If the co-ordinate is oriented in such a way that $\theta_0 = 0$, we have

$$\frac{1}{r} = \frac{mc}{L^2}(1 + \frac{AL^2}{mc} \cos(\theta)).$$

This gives the polar (r, θ) equation of the orbit of a particle moving under an inverse square law of attractive force. Comparing the above last equation with the general equation of a conic in polar co-ordinates

$$\frac{1}{r} = \frac{1}{J}(1 + \epsilon \cos(\theta)),$$

we find that it represents a conic section with $J = \frac{L^2}{mc}$ and $\epsilon = \frac{AL^2}{mc}$. The quantity ϵ is called the eccentricity. The shape of the trajectory of the particle, therefore, depends upon the value of ϵ and hence on L and A .

- (a) For $\epsilon > 1$, the path is a hyperbola.
- (b) For $\epsilon = 1$, the path is a parabola.
- (c) For $0 < \epsilon < 1$, the path is an ellipse.
- (d) For $\epsilon = 0$, the path is a circle.

12. Starting from the differential equation of motion of a particle moving under central force:

$$F(r) = F(1/u) = -\frac{L^2 u^2}{m} \left(\frac{d^2 u}{d\theta^2} + u \right)$$

obtain the polar (r, θ) equation of the orbit of a particle moving under an inverse square law of repulsive force (eg., Coulomb force) [Or, derive the equation $\frac{1}{r} = \frac{1}{J}(\epsilon \cos(\theta) - 1)$; where $J = \frac{L^2}{mc}$ and $\epsilon = \frac{AL^2}{mc}$.]

13. From the equation of the orbit of a particle moving under attractive inverse square force

$$\frac{1}{r} = \frac{mc}{L^2} \left(1 + \frac{AL^2}{mc} \cos(\theta - \theta_0) \right),$$

find the minimum and maximum value of r . [Solution: For r to be minimum or, $1/r$ to be maximum, we must have $(\theta - \theta_0) = 0, \Rightarrow \cos(\theta - \theta_0) = 1$. Hence $(\frac{1}{r})_{max} = \frac{mc}{L^2} + A$. And for r to be maximum or, $1/r$ to be minimum, we must have $(\theta - \theta_0) = \pi, \Rightarrow \cos(\theta - \theta_0) = -1$. Hence $(\frac{1}{r})_{min} = \frac{mc}{L^2} - A$.]

14. With the help of the expression for total energy E of the particle under a central force given by

$$E = \frac{1}{2}mv^2 + U + \frac{L^2}{2mr^2},$$

and considering attractive potential ($U = -c/r; c = GMm$), define turning radii. Hence obtain, expressions for A and ϵ appearing in the equation

$$\frac{1}{r} = \frac{mc}{L^2} \left(1 + \frac{AL^2}{mc} \cos(\theta) \right)$$

[Solution: For attractive potential, $U = -c/r, c = GMm$;

$$E = \frac{1}{2}mv^2 + U + \frac{L^2}{2mr^2} = \frac{1}{2}mv^2 - c/r + \frac{L^2}{2mr^2}.$$

At the turning points (where r is maximum or minimum), $v = \dot{r} = 0$. Therefore, $E = \frac{L^2}{2mr^2} - c/r \Rightarrow \frac{L^2}{2m} \left(\frac{1}{r} \right)^2 - c \left(\frac{1}{r} \right) - E = 0 \Rightarrow$ A quadratic equation in $(\frac{1}{r})$. The two roots will give $(\frac{1}{r})_{max}$ and $(\frac{1}{r})_{min}$ respectively.

General solution: $\frac{1}{r} = c \pm \frac{\sqrt{c^2 + \frac{4L^2 E}{2m}}}{\frac{L^2}{m}} \Rightarrow \frac{1}{r} = \frac{mc}{L^2} \pm [(\frac{mc}{L^2})^2 + (\frac{2mE}{L^2})]^{1/2}$.

Therefore

$$\left(\frac{1}{r} \right)_{max} = \frac{mc}{L^2} + [(\frac{mc}{L^2})^2 + (\frac{2mE}{L^2})]^{1/2}$$

and

$$\left(\frac{1}{r} \right)_{min} = \frac{mc}{L^2} - [(\frac{mc}{L^2})^2 + (\frac{2mE}{L^2})]^{1/2}$$

But we have,

$$\left(\frac{1}{r} \right)_{max} = \frac{mc}{L^2} + A$$

and

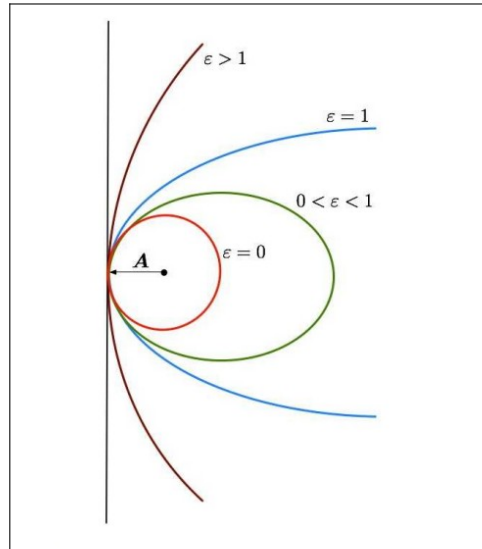
$$\left(\frac{1}{r} \right)_{min} = \frac{mc}{L^2} - A$$

Thus we have, $A = \sqrt{\left(\frac{mc}{L^2}\right)^2 + \left(\frac{2mE}{L^2}\right)} \Rightarrow A^2 = \frac{m^2c^2}{L^4} + \frac{2mE}{L^2}$. And $\epsilon = \frac{AL^2}{mc} \Rightarrow \epsilon^2 = \frac{A^2L^4}{m^2c^2} = 1 + \frac{2L^2E}{mc^2} \Rightarrow \epsilon = \sqrt{1 + \frac{2L^2E}{mc^2}}$. Thus in reference to the equation

$$\frac{1}{r} = \frac{1}{J}(1 + \epsilon \cos(\theta)),$$

for $\epsilon = \sqrt{1 + \frac{2L^2E}{mc^2}}$, we may conclude that

- (a) For $E > 0$ (+ve energy); $\epsilon > 1$, the path is a hyperbola.
- (b) For $E = 0$ (Zero energy); $\epsilon = 1$, the path is a parabola.
- (c) For $E < 0$ (-ve energy); $\epsilon < 1$, the path is an ellipse.
- (d) For $E = -\frac{mc^2}{2L^2}$; $\epsilon = 0$, the path is a circle.



Kepler's orbits are conic sections for different values of eccentricity

($\epsilon = \sqrt{1 + 2L^2E/mc^2}$). $L = mr^2\omega$, $c = GMm$, E = total energy at turning point

i) $E=0$, $\epsilon=1$, parabola ii) $E=-mc^2/2L^2$, $\epsilon=0$, circle

iii) $-mc^2/2L^2 < E < 0$, $\epsilon < 1$, ellipse iv) $E > 0$, $\epsilon > 1$, hyperbola

]

15. State and prove Kepler's laws of planetary motion. [1st law: Every planet revolves around the sun in elliptical orbit, with the sun at one of its foci. 2nd law:]
16. Derive Newton's law of gravitation from Kepler's laws.
17. The minimum and maximum separation of a comet from the sun are 7×10^{10} m and 1.4×10^{12} m respectively. If the speed of the comet at the nearest point is 6×10^4 m/s, determine the speed of the comet at the farthest position. [Ans: 3000 m] [Hints: Angular momentum conserved]
18. The moon moves in a circular orbit of 3.8×10^5 km around the earth in 27 days and the earth takes 365 days to complete one round in circular

orbit of radius 1.5×10^8 km about the Sun. Estimate mass of the Sun in terms of the mass of the earth. [Ans. $M_S \simeq 3.37 M_E$] [Hints: Gravitational attraction = centrifugal repulsion $\Rightarrow \frac{GMm}{r^2} = mr\omega^2$]

19. Identify the conservative and non-conservative forces in the following:

(a) $\vec{F} = (2xy + z^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$

(b) $\vec{F} = (y^2 - x^2)\hat{i} + 3xy\hat{j}$

(c) $\vec{F} = y\hat{i} + x\hat{j} + xy\hat{k}$

[Hints: Check whether $\nabla \times \vec{F} = 0$ or not. For $\nabla \times \vec{F} = 0$, $\vec{F}$ is conservative. For $\nabla \times \vec{F} \neq 0$, $\vec{F}$ is non-conservative.]

20. Here is an interesting problem that is a key to the nuclear reactor physics. In nuclear reactor uranium produces high energy neutrons which need to be slowed down using moderators implanted in the reactor. The first nuclear reactor built in Chicago USA 1942 used graphite (carbon) as moderators. One such neutron (mass 1.0 u) travelling at 2.6×10^7 m/s undergoes a head-on elastic collision with a carbon (graphite) nucleus (mass 12 u) initially at rest.

- Neglecting external forces during the collision, find the velocities after collision. (1 u is the atomic mass unit equaling 1.66×10^{-27} kg.). [Ans: $v_n = -2.2 \times 10^7$ m/s, $v_C = 0.4 \times 10^7$ m/s.]
- Estimate the energy transferred from the projectile (neutron) to the target (uranium nucleus) in this collision. [Ans: 28% transferred]
- What amount of energy will be transferred if the collision happens in-elastically?
- Will there be any energy transfer to the target if the neutron was at rest and uranium nucleus moving with same kinetic energy as that of the neutron (in the previous cases) collides elastically with the neutron?
- Calculate the energy transfer (if any) assuming the collision as an inelastic one. Summarize the results in tabular form, analyse them and make your own conclusion based on your findings.

21. On a frictionless table, a billiard ball (mass $m_A = 0.5$ kg) moving in the positive x-direction (velocity $u_A = 4.0$ m/s) collides elastically with another billiard ball (mass $m_B = 0.3$ kg) initially at rest (velocity $u_B = 0$). Ball A has final velocity $v_A = 2$ m/s in an unknown direction θ . Find the final velocity v_B of the ball B moving in the unknown direction ϕ . Also find the directions θ and ϕ . [Ans: $v_B = 4.47$ m/s; $\theta = 36.9^\circ$; $\phi = 26.6^\circ$] [Hint: Apply principles of conservation of momentum and kinetic energy assuming two-dimensional collision]

22. A neutron of mass 1.67×10^{-27} kg and speed 10^5 m/s collides with a stationary deuteron of mass 3.34×10^{-27} kg. The particles do not stick together and no kinetic energy is lost in collision. What are the subsequent speeds of the particles? [Ans. $v_n = 3.33 \times 10^4$ m/s, $v_d = 6.67 \times 10^4$ m/s in opposite direction]

23. Define centre of mass of a system of particles. Write down the most general

expression for the position of the centre of mass of a system of particles using vector algebra. Using the above expression write down expressions for the Cartesian components of the centre of mass.

24. How can you reduce a two-body problem into a one-body problem? Or, Derive the equation of motion of CoM of two-body system. [Solution: $R = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2}$. When there is no external force, and there is only mutual interaction. Velocity of CM will be constant. And the CM lie on the line joining the particles the force on m_1 due to m_2 as well as force on m_2 due to m_1 will act towards the centre of mass. Hence these forces are central forces. Let F_{12} is the force on m_1 due to m_2 , $F_{12} = F(r)\hat{r}$, and that on m_2 due to m_1 will be equal and opposite, $F_{21} = -F(r)\hat{r}$. $\therefore m_1 \frac{d^2 r_1}{dt^2} = F_{12} = F(r)\hat{r}$; $m_2 \frac{d^2 r_2}{dt^2} = F_{21} = -F(r)\hat{r}$; $\frac{d^2 r_1}{dt^2} - \frac{d^2 r_2}{dt^2} = (1/m_1 + 1/m_2)F(r)\hat{r} \Rightarrow \frac{d^2 r}{dt^2} = \mu F(r)\hat{r} \Rightarrow \mu \ddot{r} = F(r)\hat{r}$. Here $r = r_1 - r_2$ = relative coordinate. This is the equation of motion of CM of a two-body system. In this way two-body problem can be reduced to a one-body problem.]
25. In a water molecule, the oxygen-hydrogen separation $d = 9.57 \times 10^{-11}$ m. Each hydrogen atom has mass 1.0 u, and the oxygen atom has mass 16.0 u. Find the position of the centre of mass of the water molecule. Angle between the O-H bonds is 105° . [Ans: $(6.5 \times 10^{-12}$ m, 0 m)]
26. What is a rocket? Describe rocket motion and derive an expression for its acceleration assuming a gravity free region. [Ans. **Rocket** is a device used to take a satellite at the desired height and also to put it in a desired orbit.] [Take: absolute velocity of the rocket as v , absolute velocity of exhaust of fuel material as v_g , velocity of the exhausted fuel with respect to the rocket $v_{fuel} (= -v_g + v)$, mass of ejected fuel ($-dm$), mass of rocket ($m + (-dm)$). At time t : Momentum (mv); At time $t+dt$: total momentum $= (-dm)(v_{fuel}) + (m + dm)(v + dv)$. Conservation of linear momentum gives: $m \frac{dv}{dt} = -v_g \frac{dm}{dt}$; $\int_{v_0}^v dv = -v_g \int_{m_0}^m \frac{dm}{m}$] [**Alter:** Assuming an inertial reference frame on the earth: Let at any instant t the mass of the rocket with fuel is M and its speed is V . So, the linear momentum, $P(t) = MV$. Let at any subsequent instant $(t+dt)$, the mass of the rocket is $(M - dM)$ and its speed is $(V + dV)$, so the forward linear momentum of the rocket is $(M - dM)(V + dV)$ and the backward linear momentum of the ejected gas w.r.t. the earth is $dM(V_g + V)$. Hence, the total linear momentum of the rocket and plus ejected gas, $P(t+dt) = (M - dM)(V + dV) + dM(V + V_g)$. Therefore, the change in linear momentum, $dP = P(t + dt) - P(t) = (M - dM)(V + dV) + dM(V + V_g) - MV = MV + MdV - VdM - dMdV + VdM + V_g dM - MV = MdV + V_g dM - dMdV \simeq MdV + V_g dM \Rightarrow \frac{dP}{dt} = M \frac{dV}{dt} + V_g \frac{dM}{dt}$. According to Newton's second law of motion, if the external force acting on the rocket is F_e , then, $F_e = \frac{dP}{dt} = M \frac{dV}{dt} + V_g \frac{dM}{dt} \Rightarrow M \frac{dV}{dt} = F_e - V_g \frac{dM}{dt}$. In the absence of any external force (like the force of gravity), $F_e = 0$, and, $M \frac{dV}{dt} = -V_g \frac{dM}{dt}$. Hence, Acceleration, $a = \frac{dV}{dt} = -\frac{V_g}{M} \frac{dM}{dt}$.
27. A bomb weighing 50 kg explodes into three pieces in flight when its velocity is $20\hat{i} + 22\hat{j} + 10\hat{k}$ m/s. Two fragments of the bomb weighing 10 and 20 kg, are found to have velocities $100\hat{i} + 50\hat{j} + E = 20\hat{k}$ m/s and $30\hat{i} + 20\hat{j} + 10\hat{k}$ m/s. Find the velocity of the third piece. [Ans. $-30\hat{i} + 50\hat{j} - 25\hat{k}$ m/s.]

[Hint: Use linear momentum conservation: $M\vec{V} = m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3$. Let velocity of the third particle, $\vec{v}_3 = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$. Now, equate the components to find $\vec{v}_3$.]

28. (a) What is driven (or forced harmonic) oscillator? How it differs from the simple harmonic, and damped harmonic oscillators?
- (b) A particle of mass m is subjected to a damping force proportional to its instantaneous velocity and a driving force $F_0 \cos \omega t$. Write down the equation of motion. Solve the equation to obtain instantaneous displacement and velocity. [Hint: $m \frac{d^2x}{dt^2} + r \frac{dx}{dt} + kx = F_0 \cos \omega t \Rightarrow \frac{d^2x}{dt^2} + 2b \frac{dx}{dt} + \omega_0^2 x = f_0 \cos \omega t$; $2b = r/m$; $\omega_0 = \sqrt{k/m}$; $f_0 = F_0/m$. (A) Transient solution: $x = A_0 e^{-bt} \sin(\sqrt{\omega_0^2 - b^2}t + \phi) = A_0 e^{-bt} \sin(\omega' t + \phi)$; $\omega' = \sqrt{\omega_0^2 - b^2}$, $A_0 = \frac{\omega_0 x_0}{\sqrt{\omega_0^2 - b^2}}$, $\phi = \tan^{-1} \left(\frac{\sqrt{\omega_0^2 - b^2}}{b} \right)$ (B) Steady state solution: Let the steady state solution of $m \frac{d^2x}{dt^2} + r \frac{dx}{dt} + kx = F_0 e^{i\omega t} \Rightarrow \frac{d^2x}{dt^2} + 2b \frac{dx}{dt} + \omega_0^2 x = f_0 e^{i\omega t}$ be $x = A e^{i\omega t}$. Then, $(-\omega^2 + i\omega 2b + \omega_0^2) A e^{i\omega t} = f_0 e^{i\omega t} \Rightarrow A = \frac{f_0}{-\omega^2 + i2b\omega + \omega_0^2} = \frac{f_0}{(\omega_0^2 - \omega^2) + i(2b\omega)} = \frac{f_0((\omega_0^2 - \omega^2) - i f_0(2b\omega))}{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2} = \left(\frac{f_0(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + 4b^2\omega^2} \right) + i \left(\frac{-2bf_0\omega}{(\omega_0^2 - \omega^2)^2 + 4b^2\omega^2} \right)]$
- (c) Graphically show nature of displacement-time graph assuming large, small and critical damping.